

TYPED CONTROL, INFORMATION, AND SADDLE ENVELOPES FOR KINETIC COTANGENT PHASES

QIAN QI

ABSTRACT. Positive posterior slice measures are separated from oriented collision currents. A compact evidence-direction cone supplies the zero-evidence state, and integrated coarea disintegration gives a Feller belief kernel for definable observation strata. Inserted source-dependent coefficients, a predeclared analytic reachable-chaos class, belief dynamic programming, and an explicit controlled LAN/Bernstein–von Mises theorem follow.

Platform identifier: HSBG-CONTROL-v4. **Controlling revision:** ROUND12-REFEREE-POSITIVE-CLOSURE. **Registered proof source:** C1_POSITIVE_SLICE_EVIDENCE_CONE.tex.

CONTENTS

1. Round-twelve positive closure: positive slice measures, evidence cones, and controlled belief limits	1
1.1. Definable observation strata and positive slices	2
1.2. The compact evidence cone	2
1.3. Inserted observation coefficients	3
1.4. Reachable quantitative chaos and dynamic programming	3
1.5. An explicit posterior tangent	4
Scope and verification status	4

1. ROUND-TWELVE POSITIVE CLOSURE: POSITIVE SLICE MEASURES, EVIDENCE CONES, AND CONTROLLED BELIEF LIMITS

Oriented collision currents and positive posterior measures are separated. Evidence is the mass of a positive disintegration measure, never the pairing of a positive-dimensional current with the scalar 1. Zero evidence is handled by a compact cone over a tight set of probability directions.

Date: August 31, 2026.

2020 Mathematics Subject Classification. Primary 49L25, 93E20; Secondary 60G35, 82C40.

Key words and phrases. risk-sensitive control, kinetic equation, filtering, saddle envelope, deterministic information.

1.1. Definable observation strata and positive slices. Fix a finite decision horizon and a finite family of real-analytic, globally subanalytic observation maps O_k^u . Together with the hard-sphere collision faces they generate a locally finite Whitney stratification $\mathcal{S}_k$. On each stratum, the predicted law has a positive coefficient measure μ_S and a separate oriented normal current J_S encoding the transport/collision geometry.

Theorem 1.1 (Positive coarea disintegration). *For every regular stratum map $O : S \rightarrow \mathbb{R}^d$ there are a positive evidence measure ζ_O and positive conditional probability measures $\nu_{O,y}$ such that*

$$\int_S F d\mu_S = \int \left(\int_{O^{-1}(y)} F d\nu_{O,y} \right) d\zeta_O(y).$$

The oriented slice $\langle J_S, O, y \rangle$ is retained as an additional geometric coordinate but is not used as a posterior mass. Critical strata are disintegrated at their actual rank and inserted as lower-rank positive components.

Proof. Apply the coarea theorem to the positive rectifiable coefficient measure in each constant-rank chart, including its Jacobian in ζ_O . Rokhlin disintegration normalizes the fiber measures. Federer slicing of the oriented current supplies the independent geometric coordinate. Global subanalyticity gives a finite Whitney decomposition on each chronological cell; Hardt triviality makes the refinements locally finite through the fixed number of decisions. Summing strata proves the formula. $\square$

1.2. The compact evidence cone. Let

$$\mathcal{K}_M = \{\nu \in \mathcal{P} : \nu(W) \leq M\}$$

with the weighted weak topology; it is compact by the common Lyapunov moment bound. Define the cone

$$\text{Cone}(\mathcal{K}_M) = ([0, 1] \times \mathcal{K}_M) / (\{0\} \times \mathcal{K}_M),$$

where $r = Z/(1 + Z)$ is the evidence radius. To retain the approach direction at zero, use the real oriented blow-up $[0, 1] \times \mathcal{K}_M$ and let the transition act on both coordinates.

Theorem 1.2 (Feller evidence-direction kernel). *Prediction followed by the positive disintegration of Theorem 1.1 defines a Feller kernel on the compact blown-up state*

$$(r, \nu, (J_S)_S) \in [0, 1] \times \mathcal{K}_M \times \mathcal{J}_M.$$

At $r = 0$ the next state is determined by the retained probability direction ν and its predicted positive slice; if that slice again has zero mass, the same blow-up rule is iterated. The kernel is independent of the approximating positive-evidence sequence.

Proof. The B2 prediction semigroup preserves positivity and satisfies the common weighted Lyapunov bound. Integrated coarea disintegration is continuous for bounded continuous tests after summing the finitely many rank

strata; pointwise continuity of an individual fiber is not asserted. Normalize only when the new evidence is positive. If it tends to zero, divide before taking the limit and retain the direction in the compact set $\mathcal{K}_M$. Continuity of prediction and integrated slices makes this limit unique. Normal-current compactness with fixed stratum dimension and the chronological local-finiteness bound gives compactness of $\mathcal{J}_M$. $\square$

1.3. Inserted observation coefficients. The observation statistics used below are affine continuous functionals of the particle path and actual-contact measure. Their source is therefore a finite additional coordinate of the B2 marked pressure.

Theorem 1.3 (Source-uniform inserted coarea coefficient). *On every regular observation chart, the exact joint number–constraint– observation coefficient has the B1 relative local-limit expansion, uniformly with one bounded future insertion. Numerator and denominator use their own source-dependent saddles. The denominator is zero-free on a common complex neighborhood of the real regular chart, and their ratio converges to the conditional prepared path law.*

Proof. Adjoin the observation coordinates to the full B2 pressure, not to independent spatial blocks. The affine-span condition is the nondegeneracy of the joint pressure Hessian. B1’s compact-annulus and high-frequency pressure gaps apply to this enlarged finite vector. A bounded insertion is represented by a central marked transfer derivative and has the same Fourier error estimates. Separate saddle equations are solved for numerator and denominator. The positive real Gaussian coefficient has a uniform lower bound on the regular chart; Rouché’s theorem and the relative error exclude nearby complex zeros. Division gives the ratio theorem. $\square$

1.4. Reachable quantitative chaos and dynamic programming. Fix a complex source radius $r_* > 0$. Let $\mathfrak{R}_\varepsilon(r_*, M)$ be the set of belief laws whose factorial cumulant generating functional is analytic on that radius, has norm at most M , and whose evidence direction lies in $\mathcal{K}_M$. This class is defined before any filtering limit.

Lemma 1.4 (Invariant reachable class). *For short time and the admissible observation/control family, $\mathfrak{R}_\varepsilon(r_*, M)$ is invariant after increasing M by a fixed factor. If κ_{mfin} denotes the finitely retained correlation coordinates, then the full and reduced prediction–observation kernels admit a coupling with weighted Wasserstein error $o(1)$ uniformly over the class and over strategies.*

Proof. B2 normal source convergence and Theorem 1.3 preserve a smaller common analytic radius under prediction and Bayes division. Cauchy estimates give the factorial cumulant bounds. Exponential moment tightness makes the cumulants moment determining. Couple laws by matching the retained cumulants and use the analytic generating functional to bound the

remaining test discrepancy by the tail of its absolutely convergent series. All constants are uniform on the compact control set. $\square$

Theorem 1.5 (Belief DPP and reduced game). *The exact belief state has a Feller dynamic programming principle for one-time preparation, reward control, and adaptive canonical actual-contact control. On $\mathfrak{R}_\varepsilon(r_*, M)$ the reduced finite-correlation game value converges uniformly to the full belief value. The error after N decision stages is at most C_N times the one-step Wasserstein error.*

Proof. Ionescu–Tulcea constructs the law from the Feller belief kernel and the nonanticipative strategies. Conditioning at a decision time gives the exact DPP. Bounded Lipschitz rewards and transition kernels propagate a one-step Wasserstein error by the Bellman contraction; induction gives the displayed finite-horizon error. $\square$

1.5. An explicit posterior tangent. Let $\vartheta \in \mathbb{R}^p$ parameterize one identifiable regular phase, with a prior density continuous and positive near ϑ_0 . Assume the joint observation coefficient above has nonsingular information matrix $\mathcal{I}(\vartheta_0)$ and the parameter stays a fixed distance from phase boundaries.

Theorem 1.6 (Controlled LAN and Bernstein–von Mises). *Uniformly over admissible strategies and bounded local alternatives,*

$$\log \frac{d\mathbb{P}_{\vartheta_0+h/\sqrt{\mu}}}{d\mathbb{P}_{\vartheta_0}} = h^T \Delta_\varepsilon - \frac{1}{2} h^T \mathcal{I} h + o_{\mathbb{P}}(1), \quad \Delta_\varepsilon \Rightarrow N(0, \mathcal{I}).$$

The posterior of $\sqrt{\mu}(\vartheta - \hat{\vartheta}_\varepsilon)$ converges in total variation to $N(0, \mathcal{I}^{-1})$.

Proof. Differentiate the exact inserted coefficient at the finite saddle; B3 supplies the joint Gaussian score and its information matrix. The common complex radius gives a uniform cubic remainder. Identifiability and the global observation LDP make posterior mass outside a shrinking regular neighborhood exponentially small. On that neighborhood, change variables $h = \sqrt{\mu}(\vartheta - \vartheta_0)$ and apply dominated Laplace convergence using the positive continuous prior. $\square$

Theorem 1.7 (Round-twelve C1 closure). *The controlled hard-sphere filter has a well-typed positive sliced-measure state, compact zero-evidence blow-up, Feller DPP, inserted local coefficient, quantitative reachable-chaos reduction, and the posterior tangent of Theorem 1.6.*

Proof. Combine Theorems 1.1, 1.2, 1.3, 1.5, and 1.6. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-twelve dependency ledger. Repository source, proof-structure, counterexample, and build

gates are internal reproducibility checks and do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: `qiqian@pku.edu.cn`